

PHYSICS MANDATORY TOP 10

Most Repeated Questions (Class 12 Board Sets 2023-2025)

1. OPTICAL INSTRUMENTS 5 MARKS

Draw a labeled ray diagram for a Compound Microscope or Astronomical Telescope in normal adjustment. Derive its magnifying power.

Recurring Evidence: Appears in almost every set analyzed for long-answer sections.

2. WAVEFRONTS & DERIVATIONS 3 MARKS

Define a wavefront. Using Huygens' Principle, verify the laws of refraction (Snell's Law) or the laws of reflection.

Critical Path: Standard derivation for wavefront interaction at plane surfaces.

3. PN JUNCTION RECTIFIERS 3-5 MARKS

Explain the working of a Full Wave Rectifier with a circuit diagram and input/output waveforms.

Biasing Note: Often paired with questions on depletion layer changes during biasing.

4. BOHR'S ATOMIC MODEL 3 MARKS

Using Bohr's postulates, derive the expression for the radius of the n -th orbit or the total energy of an electron.

Check: Memorize the physical constant values (Z , e , m) used in these derivations.

5. PHOTOELECTRIC GRAPHS CASE STUDY / 2 MARKS

Analyze the graph of stopping potential vs frequency. Determine work function and threshold frequency from the slope and intercept.

Trend: Highest frequency of appearance in MCQ and Case-Study sections.

6. DRIFT VELOCITY 2-3 MARKS

Derive the relation between electric current and drift velocity. Explain how resistivity changes with relaxation time.

Logic: High chance of being asked in context of temperature effects on conductors vs semiconductors.

7. MOVING COIL GALVANOMETER 5 MARKS

Explain the principle and working of a Moving Coil Galvanometer. Show how to convert it into a Voltmeter or Ammeter.

Detail: Be sure to mention the importance of the "Radial Magnetic Field."

8. CAPACITOR & DIELECTRICS 3 MARKS

Derive the capacitance for a parallel plate capacitor containing a dielectric slab. Explain energy change with battery disconnected.

Formula Core: $C = K\epsilon_0 A/d$. Know the factor K impact on all 4 variables (Q, V, C, E).

9. NUCLEAR DENSITY 2 MARKS

Prove that the nuclear density is independent of the mass number. Calculate the ratio of densities for two different nuclei.

Proof: Final answer should always be 1:1, regardless of the elements given.

10. PRISM DERIVATIONS 3-5 MARKS

Derive the Prism Formula relating refractive index to the angle of minimum deviation (D_m) and prism angle (A).

Variation: May be combined with Lens Maker's Formula numericals.